

ANALYSTLAB AFRICA — HEALTHCONNECT EXPERIENCE LAB

Week 5: Initial HealthConnect Analytics Report — Data Analytics Track

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Part 1: Review of Week 4 Foundation

In Week 4, the problem defined was understanding which factors in the HealthConnect appointment data are associated with patient no-shows. The proposed approach was to calculate five KPIs (overall no-show rate, no-show rate by reminder status/channel, average distance/waiting time by outcome, and repeat no-show rate), each linked to one of three business questions on reminders, access factors, and repeat no-show behaviour. Key assumptions were that the dataset is representative of real clinic patterns and that missing distance/waiting-time values (1.8% and 1.2%) would not materially affect analysis. Week 5 builds directly on this foundation by calculating and interpreting those KPIs.

Part 2: Data Preparation

The dataset (5,000 rows, 18 columns) was re-inspected before analysis:

- No duplicate rows or appointment_ids were found.
- reminder_channel is recorded as the text value "None" (not a blank) for the 1,366 patients who received no reminder — this is a deliberate data value, not missing data.
- distance_to_clinic_km has 90 missing values (1.8%) and waiting_time_minutes has 60 missing values (1.2%); these rows were excluded only from the specific averages that use those columns, and were not removed from the dataset.
- All categorical values (gender, appointment_type, appointment_outcome, reminder_channel) were checked and found consistent, with no unexpected categories.

A derived field, repeat_offender, was created from previous_no_shows: patients with previous_no_shows > 0 were labelled "Has previous no-shows", and all others "No previous no-shows", to support KPI 5.

Part 3: Exploratory Data Analysis & KPI Development

KPI 1 — Overall No-Show Rate

Definition: The percentage of all scheduled appointments that ended in a no-show.

Calculation: 2,423 of 5,000 appointments = 48.5%.

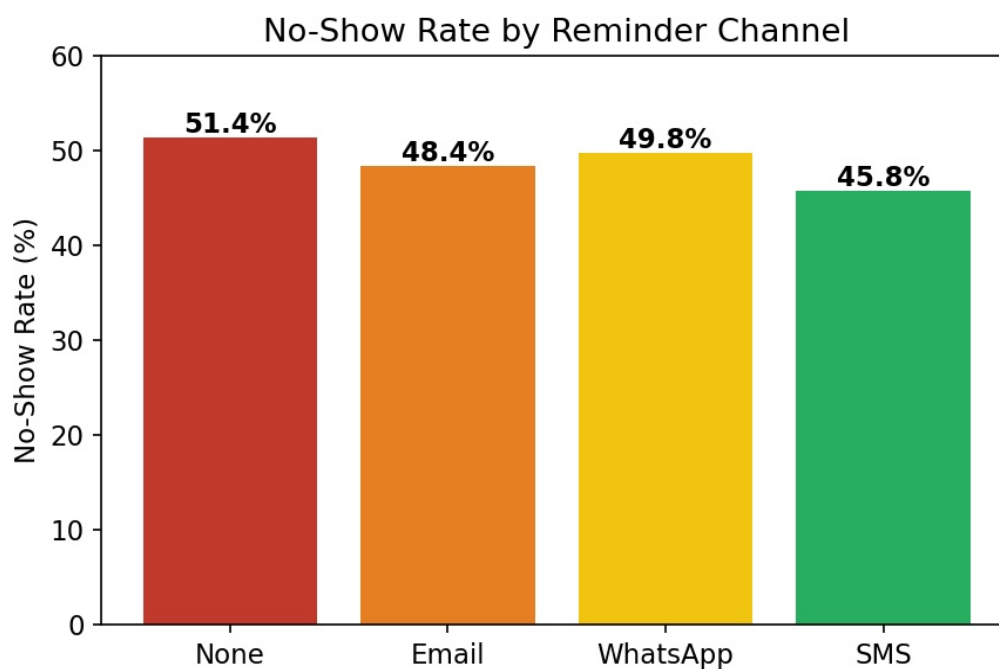
Business question: Central project question — how big is the missed-appointment problem?

Interpretation: Almost half of all appointments are missed. This confirms the scale of HealthConnect's problem and sets the baseline against which every other KPI below is compared.

KPI 2 — No-Show Rate by Reminder Channel

Definition: No-show rate broken down by which reminder channel (or none) the patient received.

Calculation: None 51.4%, WhatsApp 49.8%, Email 48.4%, SMS 45.75%.



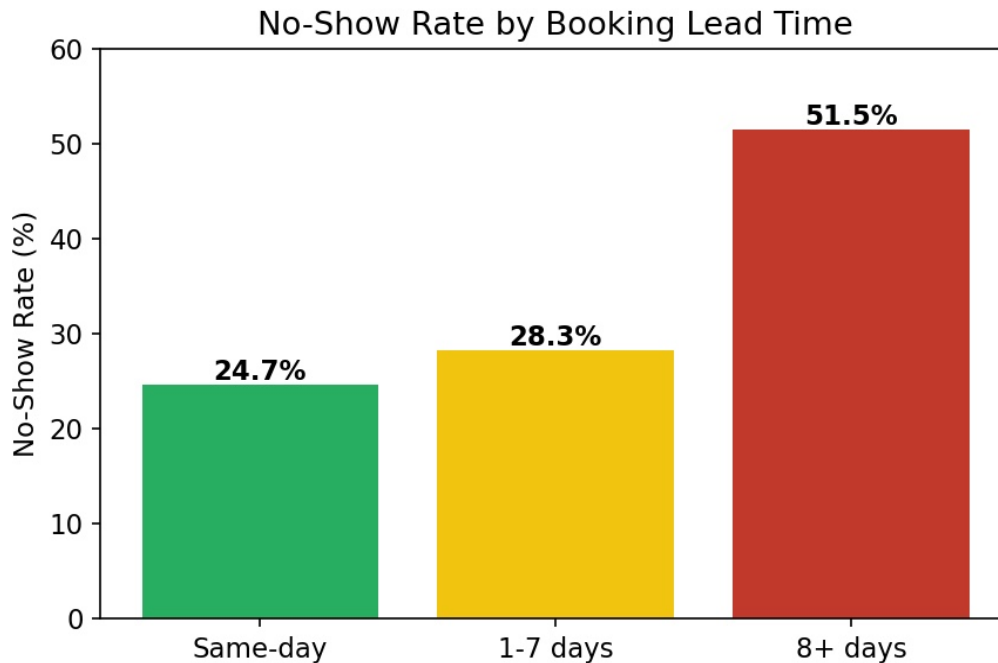
Business question: Does sending a reminder, and the channel used, reduce no-shows?

Interpretation: Any reminder is better than none, but the effect is modest — the best channel (SMS) is only about 5.7 percentage points better than sending nothing. SMS is the most effective channel tested, followed by Email, then WhatsApp.

KPI 3 — No-Show Rate by Booking Lead Time

Definition: No-show rate grouped by how far in advance the appointment was booked (same-day, 1–7 days, 8+ days).

Calculation: Same-day 24.7%, 1–7 days 28.3%, 8+ days 51.5%.



Business question: This extends the original Week 4 questions — does how far ahead an appointment is booked affect attendance?

Interpretation: This is the strongest pattern found in the whole dataset. Appointments booked 8 or more days in advance are more than twice as likely to be missed as same-day bookings. Patients likely forget or deprioritise appointments booked far ahead.

KPI 4 — Average Distance and Waiting Time by Outcome

Definition: The average distance to the clinic and average waiting time, compared across Attended, No-Show, and Cancelled appointments.

Calculation: Distance — Attended 9.67km, No-Show 10.53km, Cancelled 10.11km. Waiting time — Attended 24.3min, No-Show 24.2min, Cancelled 23.2min.

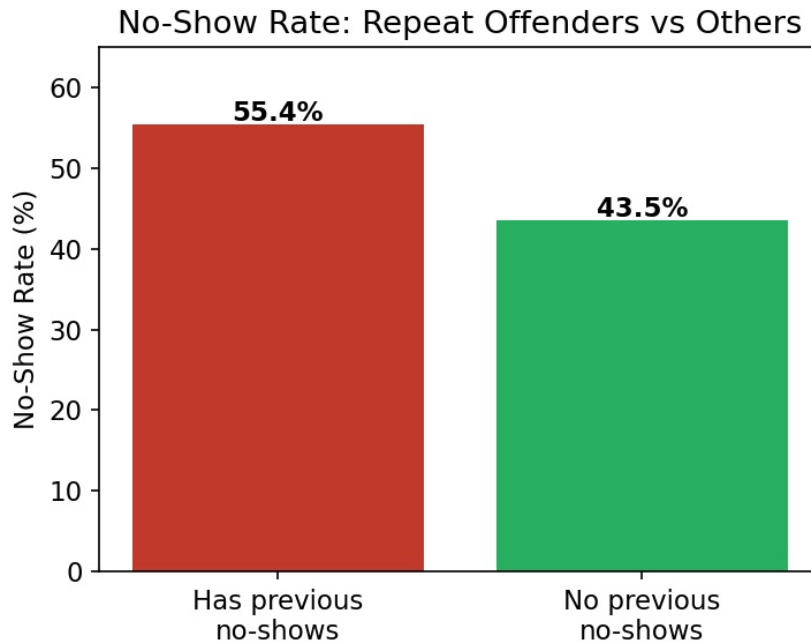
Business question: Do distance to the clinic and waiting time influence attendance?

Interpretation: Distance has only a small effect — no-show patients live, on average, less than 1km further than those who attend. Waiting time shows essentially no difference between groups. Neither factor is a strong driver of no-shows in this dataset.

KPI 5 — Repeat No-Show Rate

Definition: No-show rate for patients who have missed at least one previous appointment, compared with patients who have no history of no-shows.

Calculation: Has previous no-shows: 55.4% (1,152 of 2,079). No previous no-shows: 43.5% (1,271 of 2,921).



Business question: Are patients with a history of previous no-shows more likely to no-show again?

Interpretation: This is the second-strongest pattern in the data. Patients with any prior no-show history are almost 12 percentage points more likely to miss their next appointment too — past behaviour is a meaningful predictor of future attendance.

Part 4: Business Insights

1. Nearly half of all appointments (48.5%) end in a no-show — this is a large-scale operational problem, not a minor inefficiency, and justifies dedicated intervention.
2. Booking lead time is the strongest factor found: appointments booked 8+ days ahead have a 51.5% no-show rate versus 24.7% for same-day bookings. HealthConnect should prioritise reminder intensity for appointments booked far in advance.
3. Patients with a history of missed appointments are significantly more likely to miss again (55.4% vs 43.5%). This group could be flagged for extra reminders, confirmation calls, or overbooking strategies to offset expected no-shows.
4. SMS reminders show the lowest no-show rate (45.75%) among all channels tested, suggesting SMS should be the default or primary reminder channel where patient contact information allows.
5. Distance to clinic and waiting time show only weak or negligible relationships with no-shows, suggesting that access/logistics are not the primary driver of missed appointments — behavioural and scheduling factors (lead time, prior history) matter more.

Part 5: Assumptions, Limitations, Risks & Dependencies

- Assumption (carried from Week 4, confirmed): the dataset is representative of typical HealthConnect appointment patterns.
- Limitation: distance and waiting-time gaps (1.8%, 1.2%) were excluded from those specific averages; this is unlikely to materially change the conclusions given the small proportion affected.

- **Limitation:** reminder_channel effectiveness can only be compared for patients who received a reminder; channel choice itself may be influenced by unmeasured factors (e.g. patient preference), so the comparison is correlational.
- **Risk:** all relationships reported here are correlations, not proven causes — booking lead time and prior no-show history are strongly associated with no-shows, but a causal claim would need controlled testing.
- **Dependency:** the KPIs and insights in this report (particularly booking lead time and repeat no-show history) are strong candidates as target/feature variables for the Data Science track's predictive model.

Part 6: Cross-Track Collaboration

Track interacted with: Data Science.

Information exchanged: The KPI findings from this report — particularly that booking lead time and prior no-show history are the strongest predictors of no-shows — are directly relevant to the Data Science track's Week 5 task of defining a target variable and candidate features for a no-show prediction model.

Why relevant: The Data Analytics track's exploratory findings can shortcut the Data Science track's own feature-relevance testing, avoiding duplicated analysis across tracks.

What changed as a result: These findings are shared here so the Data Science track can consider booking_lead_days and previous_no_shows as high-priority candidate features going into their baseline model.

Part 7: Week 5 Project Summary

Planned for Week 5: Calculate the five KPIs proposed in Week 4, produce supporting visualisations, and derive at least five business insights.

Actually completed: All five KPIs were calculated and interpreted, three charts were produced, and five business insights were documented, along with an updated risks/limitations review and a cross-track collaboration note.

Key findings: Booking lead time and prior no-show history are the two strongest predictors of no-shows found in the data; reminder channel has a modest effect (SMS performing best); distance and waiting time have little to no effect.

Major challenges: Confirming exact column locations and correct text values (e.g. "None" vs blank, "No-Show" vs "No-Shows") while working in Excel required careful double-checking against the Data Dictionary.

Important decisions: Chose to treat reminder_channel = "None" as a valid category rather than missing data, since it reflects a real absence of a reminder rather than a data quality issue.

Changes to Week 4 approach: No changes — all five originally proposed KPIs remained relevant and were carried through as planned.

Cross-track collaboration completed: Shared KPI findings with the Data Science track to inform their target/feature selection (see Part 6).

Remaining work: Deeper statistical testing (e.g. correlation significance) and a full dashboard could be developed in a later phase.

Proposed focus for Week 6: Build a more complete visual dashboard summarising all KPIs, and begin exploring how these findings could translate into an intervention strategy (e.g. targeted reminder rules).